

GPS locator Communication protocol

Important revision records

- V1.0 2021-02-04 Revise the 4G device location reporting protocol. The CELLID of 4G base station needs 4 bytes
 - V1.1 2021-04-29 Modify the encoding format of 0x97 English address reply
 - V1.2 2021-05-28 Add 0x8d instruction for sending recording and transmitting large files from 0x80
 - V1.3 2021-07-29 Modify the power reporting mode (voltage level field in heartbeat package and alarm package), compatible with percentage mode
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一. Communication protocol

Introduction

This document defines the description of the application layer interface protocol of the on-board GPS locator positioning service platform. Relevant interface protocols are only applicable to the interaction between the platform and the positioning terminal.

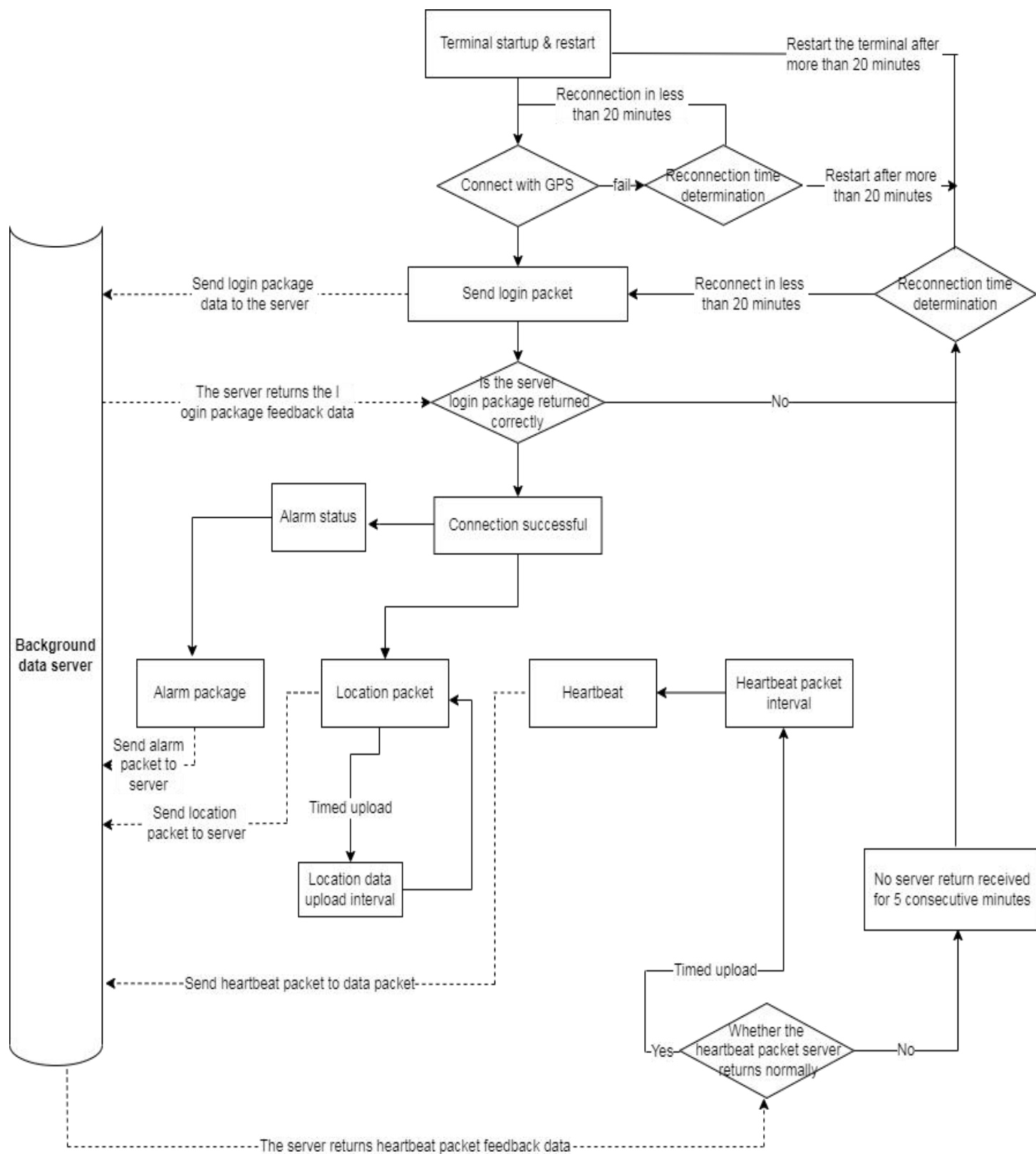
二. Terms and definitions

Terms and abbreviations	English meaning	Chinese meaning
CMPP	China Mobile Peer to Peer	中国移动点对点协议
GPS	Global Positioning System	全球卫星定位系统
GSM	Global System for Mobile Communication	全球移动通信系统
GPRS	General Packet Radio Service	通用无线分组业务
TCP	Transport Control Protocol	传输控制协议
LBS	Location Based Services	辅助定位服务
IMEI	International Mobile Equipment Identity	国际移动设备识别码
MCC	Mobile Country Code	移动用户所属国家代号
MNC	Mobile Network Code	移动网号码
LAC	Location Area Code	位置区码
Cell ID	Cell Tower ID	移动基站
UDP	User Datagram Protocol	用户数据报协议
SOS	Save Our Ship/Save Our Souls	遇难求救信号
CRC	Cyclic Redundancy Check	循环冗余校验
NITZ	Network Identity and Time Zone,	时区
GIS	Geographic Information System	地理信息系统

三. Basic rules

1. The GPRS connection is successfully established and the first login information packet is sent to the server. If the server response data packet is received within 5 seconds, the connection is considered normal, and the positioning information (GPS, LBS information packet) will be sent. After 3 minutes, the status information packet will be sent to confirm the normal communication regularly.
2. When the GPRS connection is not established successfully, the terminal cannot send the login information packet. When the GPRS connection fails for 3 times, the terminal starts the timed restart function for 20 minutes. Within 20 minutes, if the terminal successfully establishes a connection with the server and receives the data packet that the server responds to the login information packet sent by the terminal, the timed restart function will be turned off and the terminal will not restart, otherwise the terminal will restart automatically after 20 minutes.
3. After receiving the login message sent by the terminal, the server returns the response packet to the terminal. If the terminal does not receive the server return packet for more than 5 seconds after sending the login message packet or status message packet, it is considered that the current connection establishment is abnormal, start the supplementary transmission function of GPS positioning data, disconnect the current GPRS connection, re-establish a new GPRS connection and send the login message packet.
4. The connection is judged to be abnormal. If the login information packet or status information packet sent after the connection is established for three times cannot receive the data packet responded by the server, the terminal starts the timed restart function for 10 minutes. If the terminal successfully establishes the connection with the server and receives the data packet responded by the server within 10 minutes, the timed restart function is closed and the terminal does not restart, Otherwise, the terminal will restart automatically after 10 minutes.
5. After the connection is established normally, after the GPS information is changed, the terminal regularly sends the GPS and LBS combined information package to the server, and the server can set the default sending protocol through instructions.
6. In order to ensure the validity of the connection, the status information is sent to the server at fixed intervals, and the server returns the response packet confirmation.
7. For terminals without registered IMEI number, please reply to the login request response and heartbeat packet response. Do not disconnect directly. (if you directly disconnect or do not reply, the terminal will be continuously reconnected and GPRS traffic consumption will be serious).

Data flow chart



四. Packet format

The communication transmission is asynchronous and in bytes.

Total package length:

format	length(Byte)
Start bit	2
Package length	1(2)
Agreement number	1
information content	N
Information serial number	2
Error check	2
Stop bit	2

4.1. Start Bite

Fixed value, unified as hexadecimal 0x78 0x78 (packet length 1 Byte) or 0x79 0x79 (packet length 2 Byte)

4.2. Package length

Length = agreement number + information content + information serial number + error verification,

(5 + n) bytes in total, because the information content is an indefinite length field.

4.3. Agreement number

type	data
Terminal login	0x01
Login response	0x01
Location data (UTC)	0x31
Alarm data (UTC)	0x32
Alarm data reply	0x26
Chinese alarm address issued by the server	0x17
English alarm address issued by the server	0x97
Heartbeat packet (status information)	0x13
Heartbeat packet (status information) response	0x13
Lbs information (UTC)	0x34
WiFi packet	0x33
General packet for information transmission	0x94
Message transmission general packet reply	0x94
Large file transfer	0x8D
The server sends instruction information to the terminal	0x80
Command information response	0x21
GPS, telephone number inquiry address information (UTC)	0x2A
Chinese address reply	0x17
English address reply	0x97

4.4. Information content

According to different applications, corresponding to the corresponding "agreement number", determine the specific content.

4.5. Information serial number

The serial number of the first GPRS data (including status packets and GPS, LBS and other data packets) sent after power on is ' 1 ', and the serial number of each data (including status packets and GPS, LBS and other data packets) sent after power on is automatically increased by 1.

4.6. Error check

The terminal or server can use the check code to judge whether the received information is wrong. In order to prevent errors in data transmission, error verification is added to prevent data misoperation, which increases the safety and efficiency of the system. The error check code adopts crc-itu check method.

Crc-itu value of data from "package length" to "information serial number" (including "package length" and "information serial number") in the protocol body.

If the receiving party has CRC error in the calculation of the received information, it will ignore and discard the packet.

4.7. Stop bit

Fixed value, unified as hexadecimal 0x0D 0x0a.

五. Detailed explanation of sending data packet from terminal to server

Explain commonly used packet sending and server return separately

5.1. Login packet (0x01)

5.1.1. The terminal sends data packets to the server

The login information package is used to confirm the normal establishment of the connection to the server and submit the terminal ID to the server.

	format	length	Examples
Login packet (18 Byte)	Start bit	2	0x78 0x78
	Package length	1	0x11
	Agreement number	1	0x01
	Terminal ID	8	0x01 0x23 0x45 0x67 0x89 0x01 0x23 0x45
	Type identification code	2	0x00 0x00
	Time zone language	2	0x32 0x00
	Information serial number	2	0x00 0x01
	Error check	2	0x90 0x01
	Stop bit	2	0x0d 0x0a

5.1.1.1. Start bit

See packet format 4.1 for details

5.1.1.2. Package length

See packet format 4.2 for details

5.1.1.3. Agreement number

See packet format 4.3 for details

5.1.1.4. Terminal ID

BCD code with "0" before terminal ID (15 bits).

For example: 123456789012345,

Then the terminal ID is: 0x01 0x23 0x45 0x67 0x89 0x01 0x23 0x45

5.1.1.5. Type identification code

The type identification code takes up two bytes. Judge the terminal type according to this identification code.

G17H 0x00 0x00

G28 0x28 0x01

G29 0x29 0x01

5.1.1.6. Time zone language

One half byte bit15 —bit4	15	The time zone is expanded by 100	
	14		
	13		
	12		
	11		
	10		
	9		
	8		
	7		
	6		
	5		
	4		
Lower half byte bit4-bit0	3	Time zone East or west	
	2	No definition	
	1	Language selection bit	English set 1
	0	Language selection bit	Chinese set 0

Bit3 0-----Eastern time zone

1-----West time zone

If: extension: 0x32, 0x00 means East Zone 8, GMT + 8:00, Chinese.

Calculation method: $8 * 100 = 800$, converted to hexadecimal, 0x0320

Extension bit: 0x4d 0xd8 indicates west zone 12 and 3 / 4 time zone, gmt-12:45, English.

Calculation method: $12.45 * 100 = 1245$, convert to hexadecimal, 0x04,0xdd

The algorithm here is to rotate the calculated time zone value to the left by four bits, then combine the East and west of the time zone, and select the language bits, so as to save four bytes.

5.1.1.7. Information serial number

See packet format 4.5for details

5.1.1.8. Error check

See packet format 4.6 for details

5.1.1.9. Stop bit

See packet format 4.7 for details

5.1.2. Server response login packet

explain		length	Examples
Login packet (18 Byte)	Start bit	2	0x78 0x78
	Package length	1	0x05
	Agreement number	1	0x01
	Information serial number	2	0x00 0x01
	Error check	2	0xd9 0xdc
	Stop bit	2	0x0d 0x0a

Response package from the server to the terminal: (the protocol number in the response package is the same as the protocol number of the data package sent by the terminal)

5.1.2.1. Start bit

See packet format 4.1 for details

5.1.2.2. Package length

See packet format 4.2for details

5.1.2.3. Agreement number

See packet format 4.3 for details

5.1.2.4. Information serial number

See packet format 4.5for details

5.1.2.5. Error check

See packet format 4.6 for details

5.1.2.6. Stop bit

See packet format 4.7for details

5.2. Positioning data package (GPS, LBS combined information package) (0x31)

5.2.1. Terminal sends positioning data packet

format			Length (Byte)
information content	Start bit		2
	Package length		1
	Agreement number		1
	GPS information	Date time	6
		Number of GPS information satellites	1
		latitude	4
		longitude	4
		speed	1
		Heading and status	2
		Heading and status	2
	Lbs information	MCC	2
		MNC	2
		LAC	2
		Cell ID	4
	ACC		1
	Data reporting mode		1
	GPS real-time supplementary transmission		1
	serial number		2
	Error check		2
	End bit		2

5.2.1.1. Start bit

See packet format 4.1 for details

5.2.1.2. Package length

See packet format 4.2for details

5.2.1.3. Agreement number

Fixed to 0x31, see packet format 4.3 for details

5.2.1.4. Date time

(Note: all dates and times used in this document are UTC time, the same below)

format	Length (Byte)	Examples
year	1	0x0A

month	1	0x03
day	1	0x17
hour	1	0x0F
minute	1	0x32
second	1	0x17

For example: 15:50:23 on March 23, 2010

Calculation method: 10 (hexadecimal) = 0A (hexadecimal)

3 (hexadecimal) = 03 (hexadecimal)

23 (hexadecimal) = 17 (hexadecimal)

15 (hexadecimal) = 0f (hexadecimal)

50 (hexadecimal) = 32 (hexadecimal)

23 (hexadecimal) = 17 (hexadecimal)

Then the value is: 0x0a 0x03 0x17 0x0f 0x32 0x17

5.2.1.5. Length of GPS information and number of satellites involved in positioning

1byte has two hexadecimal characters. The first character is the length of GPS information(number of GPS satellites - > heading, status length) and the second character is the number of satellites involved in positioning

Example: when the value is 0xcb, it means that the length of GPS information is 12 and the number of satellites participating in positioning is 11.

(C = 12bit length, B = 11 satellites)

5.2.1.6. latitude

It occupies 4 bytes and represents the latitude value of positioning data. The numerical range is 0 to 162000000, which means the range from 0 degrees to 90 degrees. The conversion method is as follows:

Convert the longitude and latitude value output by the GPS module into a decimal in minutes; Then multiply the converted decimal by 30000 and convert the multiplication result into hexadecimal number.

For example, 22 ° 32.7658 ' = (22x60 + 32.7658) x30000 = 40582974, and then convert it to hexadecimal number

40582974 (HEX) = 26b3f3e (HEX)

The final value is 0x02 0x6b 0x3f 0x3e.

5.2.1.7. longitude

It occupies 4 bytes and represents the longitude value of positioning data. Values range from 0 to 324000000, representing a range of 0 degrees to 180 degrees.

The conversion method is consistent with that of latitude

5.2.1.8. speed

It takes up 1 byte, indicating the running speed of GPS. The value range is 0x00 ~ 0xff, indicating the range is 0 ~ 255 km / h.

For example: 0x00 stands for 0 km / h.

0x10 stands for 16 km / h

0xff stands for 255 km / h

5.2.1.9. State heading

It takes up 2 bytes, indicating the running direction of GPS, indicating the range of 0 ~ 360, unit: degrees, with due north as 0 degrees, clockwise.

BYTE_1	Bit7	0
	Bit6	0
	Bit5	GPS real-time / differential positioning
	Bit4	Has GPS positioning been
	Bit3	East longitude and west longitude
	Bit2	South latitude and north latitude
	Bit1	course
	Bit0	
BYTE_2	Bit7	
	Bit6	
	Bit5	
	Bit4	
	Bit3	
	Bit2	
	Bit1	
	Bit0	

Note: the status information in the data packet is the status at the moment recorded by the time bit in the data packet.

For example, if the value is 0x15 0x4c, it becomes binary and 00010101 01001100,

BYTE_1 Bit7 0

BYTE_1 Bit6 0

BYTE_1 Bit5 0 (real time GPS)

BYTE_1 Bit4 1 (GPS located)

BYTE_1 Bit3 0 (east longitude)

BYTE_1 Bit2 1 (north latitude)

BYTE_1 Bit1 0

BYTE_1 Bit0 1

BYTE_2 Bit7 0

BYTE_2 Bit6 1

BYTE_2 Bit5 0 → Heading 332 ° (0101001100 binary to decimal 332)

BYTE_2 Bit4 0

BYTE_2 Bit3 1

BYTE_2 Bit2	1
BYTE_2 Bit1	0
BYTE_2 Bit0	0

It means that GPS has been positioned, real-time GPS, north latitude, east longitude and heading 332 ° .

5.2.1.10. MCC

Mobile country code (MCC)

For example, China's mobile country number is: China's mobile country number is 460 (decimal) 0x01 0xCC (decimal 460 is converted to hexadecimal, and 0 is added to the left if hexadecimal is less than four digits)

The value range here is 0x0000 ~ 0x03e7

5.2.1.11. MNC

Mobile network code (MNC)

For example: 0x00 for China Mobile.

5.2.1.12. LAC

Location area code (LAC) is contained in Lai, which is composed of two bytes and adopts hexadecimal coding. The available range is 0x0001-0xffff, and code groups 0x0000 and 0xffff cannot be used (see GSM specifications 03.03, 04.08 and 11.11).

5.2.1.13. Cell ID

The cell tower ID of the mobile base station (cell ID) ranges from 0x000000 to 0xffffffff

5.2.1.14. ACC

Acc status ACC low is 00 and ACC high is 01

5.2.1.15. Data reporting mode

Reserved, no effect at present

5.2.1.16. GPS real-time supplementary transmission

0x00 real time upload

0x01 supplementary transmission

5.2.1.17. Information serial number

See packet format 4.5 for details

5.2.1.18. Error check

See packet format 4.6 for details

5.2.1.19. Stop bit

See packet format 4.7for details

5.3. Alarm package (GPS, LBS, status merge information package) (0x32)

5.3.1. The terminal sends an alarm data packet to the server

format		length (Byte)	Examples
information content	Start bit	2	0x7878
	Package length	1	0x2A
	Agreement number	1	0x32
	Date time	6	0x15051B0C000 0
	GPS information	Number of GPS information satellites	1 0xCB
		latitude	4 0x026B3F3E
		longitude	4 0x0C269CC4
		speed	1 0x78
		Heading and status	2 0x154C
	Lbs information	Lbs length	1 0x0A
		MCC	2 0x01CC
		MNC	2 0x0000
		LAC	2 0x2633
		Cell ID	4 0x02c88801
	status information	Terminal information content	1 0x44
		Voltage level	1 0x06
		GSM signal strength	1 0x03
		Alarm / language / extension port status	2 0x1301
	serial number		2 0x0003
	Error check		2 0x6A45
	End bit		2 0x0D0A

The alarm package adds lbs length and status information (alarm information) on the basis of the positioning package.

5.3.1.1. Start bit

See packet format 4.1 for details

5.3.1.2. Package length

See packet format 4.2 for details

5.3.1.3. Agreement number

Fixed to 0x32. See packet format 4.3 for details

5.3.1.4. Date time

See 5.2.1.4 format of positioning data package for details

5.3.1.5. Number of GPS information satellites

See 5.2.1.5 format of positioning data package for details

5.3.1.6. latitude

See 5.2.1.6 format of positioning data package for details

5.3.1.7. longitude

See 5.2.1.7 format of positioning data package for details

5.3.1.8. speed

See 5.2.1.8 format of positioning data package for details

5.3.1.9. Heading and status

See 5.2.1.9 format of positioning data package for details

5.3.1.11. LBS 长度

MCC - > cell ID length

5.3.1.12. MCC

See 5.2.1.10 format of positioning data package for details

5.3.1.13. MNC

See 5.2.1.11 format of positioning data package for details

5.3.1.14. LAC

See 5.2.1.12 format of positioning data package for details

5.3.1.15. Cell ID

See 5.2.1.13 format of positioning data package for details

5.3.1.16. Terminal information

One byte is used to represent various status information of the terminal.

position		Code meaning
BYTE	Bit7	1: Oil and electricity disconnection
		0: oil and electricity are connected
	Bit6	1: GPS located
		0: GPS not positioned
	Bit3~Bit5	100: SOS distress (not supported)
		011: low power alarm
		010: power failure alarm (not supported)
		001: vibration alarm (not supported)
		000: normal
	Bit2	1: Connected power charging
		0: charging without power supply
	Bit1	1: ACC high
		0: ACC low
	Bit0	1: Fortification
		0: disarm

For example: 0x44, the corresponding binary is 01000100

Indicates that the terminal status is: oil and power on, GPS has been positioned, normal without alarm, power on and charging, ACC is low and disarmed

5.3.1.17. Voltage level

(1) Battery powered

If the highest bit (bit7) is 1, it means that the power is reported as a percentage, and

the actual power percentage is equal to the value of removing the highest bit, such as 0xd3 83%.

If the highest bit (bit7) is 0, it means that the power is reported by level, ranging from 0 to 6, indicating the voltage from low to high.

0: no power (shutdown)

1: Very low power (not enough to make phone calls, send text messages, etc.)

2: Low power (low power alarm)

3: Low power (can be used normally)

4: Electric quantity

5: High power

6: Extremely high power

For example: 0x02 low power alarm sent

(2) Connected to external power

This bit is 0xf and F is a fixed value, representing the external power state; 10: ((external voltage * 10) > > 8) & 0xf

5.3.1.18. GSM signal strength level

如: 0x03 GSM Good signal 0x00: no signal;

0x01: extremely weak signal

0x02: weak signal

0x03: good signal

0x04: strong signal

For example: 0x03 GSM signal is good

5.3.1.19. Alarm / language

Front position: terminal alarm status (applicable to alarm package and requires electronic fence function items)

Last bit: the current language bit of the terminal

Byte 1	0x00: normal
	0x01: SOS for help
	0x02: power failure alarm
	0x03: vibration alarm
	0x04: fence entry alarm
	0x05: out of fence alarm
	0x06 over speed alarm
	0x09 displacement alarm
	0x0a entering GPS blind zone alarm
	0x0B out of GPS blind zone alarm
	0x0C startup alarm
	0x0E low external power alarm
	0x0Flow point protection alarm of external power
	0x11 shutdown alarm
	0x13(remove alarm)
	0x14 door alarm

	0X15 low power shutdown
	0X2C collision alarm
	0x2D flip alarm
	0x2E sharp turn alarm
	0x28 rapid deceleration alarm
	0X29 rapid acceleration alarm
Byte2	0x01 Chinese
	0x02 English

For example:

No alarm Chinese: 0x00 0x01; No alarm English: 0x00 0x02

In order to increase the reliability of alarm information, the alarm information is marked repeatedly. In most cases, this alarm information is consistent with the upper terminal information. The inconsistency is as follows:

A. Low power alarm in terminal information B. alarm / language information in and out of fence alarm

5.3.1.20. Information serial number

See packet format 4.5 for details

5.3.1.21. Error check

See packet format 4.6 for details

5.3.1.22. Stop bit

See packet format 4.7 for details

Note: the status information in the data packet is the status at the moment recorded by the time bit in the data packet.

5.3.2. The server sends an alarm packet reply to the terminal (0x26)

	format	length(Byte)	Examples
information content	Start bit	2	0x78 0x78
	Package length	1	0x05
	Agreement number	1	0x26
	serial number	2	0x01 0x33
	Error check	2	0x5d 0xab
	End bit	2	0x0d 0x0a

5.3.2.1. Start bit

See packet format 4.1 for details

5.3.2.2. Package length

See packet format 4.2 for details

5.3.2.3. Agreement number

See packet format 4.3 for details

5.3.2.4. serial number

See packet format 4.5 for details

5.3.2.5. Error check

See packet format 4.6 for details

5.3.2.6. End bit

See packet format 4.7 for details

5.3.3. The server sends an alarm data address packet reply to the terminal**5.3.3.1. Chinese reply**

The Chinese reply packet is as follows:

Instruction package sent by the server to the terminal (15 + m + n byte)	Start bit		2	示例		
	Data bit length		1	0x7878		
	Agreement number		1	45+M		
	information content	Instruction length		1	0x17	
		Server flag bit		4	39+M	
		Instruction content	ALARMSMS		8	0x00000001
			&&		2	0x414C41524D534D53
			Address content		M	0x2626
			&&		2	0x624059044F4D7F...
			Telephone number		21	0x2626
		##		2	0x00000000...	
	Information serial number		2	0x2323		
	Check bit		2	0x0004		
Stop bit		2	0x6A3E			

Request Chinese address reply agreement No.: 0x17

Instruction content: alarmsms & & address content & & phone number (all 0) ##
(alarmsms, & &, ## are fixed strings)

The Chinese address content is distributed in Unicode.

Example of reply to Chinese address information:

Example of reply to Chinese address information:

```

seven thousand eight hundred and seventy-eight    // Start bit
eighty-five    // Data length
seventeen    // Reply agreement No
7E    // The instruction length is the length of the transmitted content
information
00000001    // Server flag bit
414C41524D534D53    // ALARMSMS
two thousand six hundred and twenty-six    //&& Separator
624059044f4d7f6e0028 / / send the Chinese position in Unicode
004C004200530029003A
5E7F4E1C77015E7F5DDE
5E0282B190FD533AFF17
FF15FF144E6190530028
004E00320033002E0033

```

Considering the long English or other foreign addresses, one byte of packet length is not enough, and it is increased to 2 bytes (the same below).

Instruction package sent by the server to the terminal (15 + m + n byte)	Start bit		2	示例	
	Data bit length		2	0x7979	
	Agreement number		1	46+M	
	informati on content	Instruction length		2	0x97
		Server flag bit		4	39+M
		Instruction content	ALARMSMS	8	0x00000001
			&&	2	0x414C41524D534D53
			Address content	M	0x2626
			&&	2	0x534F53284C29...
			Telephone number	21	0x2626
		##	2	0x00000000...	
	Information serial number		2	0x2323	
Check bit		2	0x0005		
Stop bit		2	0x5AB1		

Instruction content: alarmsms & & address content & & phone number (all 0) ##
(alarmsms, & &, ## are fixed strings)

```
7979 // start bit
0080 // data length
97 // reply to agreement No
0078 // the instruction length is the length of the sent content information
00000001 // server flag bit
414C41524D534D53 // ALARMSMS
2626 // & separator
534f53284c293a536869 // the English position is sent in ASCII
6D696E2046616972796C
616E6420576573742052
```

```
642C4875696368656E67
2C4875697A686F752C47
75616E67646F6E67284E
32332E3131312C453131
342E343131294E656172
six thousand two hundred and seventy-nine
2626 / / & separator
00000000000000000000000000000000000000000000000000000000000000000000 / / phone number
2323 / / ## content information Terminator
0007 / / serial number
72b5 / / check bit
0d0a / / stop bit
```

Note: since some alarm functions do not require the platform to reply to the address information, the platform does not need to reply to the analytical address after receiving the terminal alarm package. The types of alarms that do not need to reply to the address are as follows:

1. Low power alarm 2 Over speed alarm 3 GPS blind spot

5.4. Heartbeat packet (status packet)

Heartbeat packet is a data packet that maintains the connection between the terminal and the server

5.4.1. The terminal sends heartbeat packets to the server

format			Length (byte)	Examples
informatio n content	Start bit		2	0x78 0x78
	Package length		1	0x0a
	Agreement number		1	0x13
	status information	Terminal information content	1	0x04
		Voltage level	1	0xf0
		GSM signal strength	1	0x04
		Language / extension status	2	0x8d 0x01
	serial number		2	0x00 0x7f
	Error check		2	0x71 0x7f
End bit		2	0x0d 0x0a	

5.4.1.1. Start bit

See packet format 4.1 for details

5.4.1.2. Package length

See packet format 4.2 for details

5.4.1.3. Agreement number

0x13

5.4.1.4. Terminal information content

One byte is used to represent various status information of the terminal.

位		Code meaning
BYTE	Bit7	1: Oil and electricity disconnection
		0: oil and electricity are connected
	Bit6	1: GPS located
		0: GPS not positioned
	Bit3~Bit5	100: SOS distress (not supported)
		011: low power alarm
		010: power failure alarm (not supported)
		001: vibration alarm (not supported)
		000: normal
	Bit2	1: Connected power charging
		0: charging without power supply
	Bit1	1: ACC high
		0: ACC low
	Bit0	1: Fortification
		0: disarm

For example: 0x44, the corresponding binary is 01000100

Indicates that the terminal status is: oil and electricity are connected, GPS has been positioned, power has been connected and charged, ACC is low and disarmed

5.4.1.5. Voltage level

(1) Battery powered

If the highest bit (bit7) is 1, it means that the power is reported as a percentage, and the actual power percentage is equal to the value of removing the highest bit, such as 0xd3 83%.

If the highest bit (bit7) is 0, it means that the power is reported by level, ranging from 0 to 6, indicating the voltage from low to high.

0: no power (shutdown)

1: Very low power (not enough to make phone calls, send text messages, etc.)

2: Low power (low power alarm)

3: Low power (can be used normally)

4: Electric quantity

5: High power

6: Extremely high power

For example: 0x02 low power alarm sent

(2) Connected to external power

This bit is 0xfx and F is a fixed value, representing the external power state;X:((external voltage * 10) > > 8) & 0x0f

5.4.1.6. GSM signal strength

0x00: no signal;

0x01: extremely weak signal

0x02: weak signal

0x03: good signal

0x04: strong signal

For example: 0x03 GSM signal is good

5.4.1.7. Language / extension status

Front position: terminal expansion port status (external voltage * 10 byte0)

Example of external voltage analysis

0xf3 (voltage level field) 0x20 (expansion port status field) takes 0x320 to decimal 800 external voltage, i.e. 80.0v

5.4.1.8. serial number

See packet format 4.5 for details

5.4.1.9. Error check

See packet format 4.6 for details

5.4.1.10. End bit

See packet format 4.7 for details

5.4.2. Server response packet

format		length	示例
Heartbeat packet (18 byte)	Start bit	2	0x7878
	Package length	1	0x05
	Agreement number	1	0x13
	Information serial number	2	0x0011
	Error check	2	0xF970
	Stop bit	2	0x0D0A

Response package from the server to the terminal: (the protocol number in the response package is the same as the protocol number of the data package sent by the terminal)

5.4.2.1. Start bit

See packet format 4.1 for details

5.4.2.2. Package length

See packet format 4.2 for details

5.4.2.3. Agreement number

0x13

5.4.2.4. Information serial number

See packet format 4.5 for details

See packet format 4.6for details

5.4.2.5. Stop bit

See packet format 4.7 for details

5.5. Base station packet (LBS only) (0x34)

5.5.1. The terminal sends lbs positioning data packet

format				length(Byte)	Examples	
informa tion content	Start bit			2	0x7878	
	Package length			1	0x37	
	Agreement number			1	0x34	
	Date time			6	0x15051B0C0000	
	Lbs informat ion	TA		1	0x00	
		MCC		2	0x01CC	
		MNC		2	0x0000	
		CellNum		1	0x01	
		Base station 1	LAC	2	0x2633	
			Cell ID	4	0x02c88801	
			RSSI	1	0x4E	
		Base station2	LAC	2	0x0000	
			Cell ID	4	0x00000000	
			RSSI	1	0x00	
		Base station 3	LAC	2	0x0000	
			Cell ID	4	0x00000000	
			RSSI	1	0x00	
		Base station 4	LAC	2	0x0000	
			Cell ID	4	0x00000000	
			RSSI	1	0x00	
		Base station 5	LAC	2	0x0000	
			Cell ID	4	0x00000000	
			RSSI	1	0x00	
		reserve			3	0x000000
		serial number			2	0x0007
	Error check			2	0x2F32	
	End bit			2	0x0D0A	

5.5.1.1. Start bit

See packet format 4.1 for details

5.5.1.2. Package length

See packet format 4.2 for details

5.5.1.3. Agreement number

Fixed to 0x34. See packet format 4.3 for details

5.5.1.4. Date time

format	Length	example
--------	--------	---------

	(Byte)	
year	1	0x0A
month	1	0x03
day	1	0x17
hour	1	0x0F
minute	1	0x32
second	1	0x17

For example: 15:50:23 on March 23, 2010

Calculation method: 10 (hexadecimal) = 0A (hexadecimal)

3 (hexadecimal) = 03 (hexadecimal)

23 (hexadecimal) = 17 (hexadecimal)

15 (hexadecimal) = 0f (hexadecimal)

50 (hexadecimal) = 32 (hexadecimal)

23 (hexadecimal) = 17 (hexadecimal)

Then the value is: 0x0a 0x03 0x17 0x0f 0x32 0x17

5.5.1.5. MCC

Mobile country code (MCC)

For example, China's mobile country number is: China's mobile country number is 460 (decimal) 0x01 0xCC (decimal 460 is converted to hexadecimal, and 0 is added to the left if hexadecimal is less than four digits)

The value range here is 0x0000 ~ 0x03e7

5.5.1.6. MNC

Mobile network code (MNC)

For example: 0x00 for China Mobile.

5.5.1.7. CellNum

Actual number of base stations (Note: in the example, there is only one actual base station, and other base stations 2 - > base stations 5 need to fill in 0)

5.5.1.8. LAC

Location area code (LAC) is contained in Lai, which is composed of two bytes and adopts hexadecimal coding. The available range is 0x0001-0xffff, and code groups 0x0000 and 0xFFFF cannot be used (see GSM specifications 03.03, 04.08 and 11.11).

5.5.1.9. Cell ID

The cell tower ID of the mobile base station (cell ID) ranges from 0x000000 to 0xFFFFFFFF

5.5.1.9. RSSI

The signal strength of the main cell ranges from 0x00 to 0xff. The actual signal strength is negative. Upload its absolute value.

See Glossary - RSSI.

5.5.1.10. Information serial number

See packet format 4.5 for details

5.5.1.11. Error check

See packet format 4.6 for details

5.5.1.12. stop bit

See packet format 4.7 for details

5.6. WIFI Information package

format		length (Byte)	Explain in detail	示例
Start bit		2	0x78 0x78	0x7878
Package length		1	Length = protocol number + information content + information serial number + error verification	
Agreement number		1	0x33	0x33
information content	Date time	6	Year (1byte) month (1byte) day (1byte) hour (1byte) minute (1byte) second (1byte) (converted to decimal)	0x15051B0C0000
	MCC	2	Country code of mobile user mobile country code Mobile network code	0x01CC
	MNC	2	Mobile network code	0x0000
	LAC	2		0x2633
	CI	4	Cell Tower ID (Cell ID)	0x02c88801
	RSSI	1	The cell signal strength ranges from 0x00 to 0xff. The 0x00 signal is the weakest and the 0xff signal is the strongest.	0x4E
	NLAC1	2	Ditto LAC	0x0000
	NCI1	4	Ditto CI	0x00000000
	NRSSI1	1	Ditto RSSI	0x00
	...	...	...	
	NLAC5	2	Ditto LAC	0x0000
	NCI5	4	Ditto CI	0x00000000
	NRSSI5	1	Ditto RSSI	0x00
	Time advance	1	It refers to the difference between the actual time when the mobile station signal reaches the base station and the time when the mobile station signal reaches the base station assuming that the distance between the mobile station and the base station is 0.	0x00

GPS locator communication protocol

	WiFi quantity	1	Used to determine the number of WiFi transmitted in the packet. 0 is no WiFi detected	0x05
	WIFI MAC1	6	MAC of received signal 1 WiFi (transmit according to the actual number of WiFi searched, for example, one transmission is found, multiple transmissions are found, for example, 0 is found)	0x01A2B50E3286
	WIFI strength 1	1	Signal strength of signal 1 WiFi	0x42
	...	...	...	
	WIFI MAC5	6	ditto	0x3E0D235A096F
	WIFI strength 5	1	ditto	0x4F
Information serial number		2	After starting up, the serial number of data sent each time will be automatically increased by 1	0x000A
Error check		2	Crc-itu value from "package length" to "information serial number". If the receiving party has CRC error in the calculation of the received information, it will ignore and discard the packet	0x3F8A
Stop bit		2	Fixed value, unified as 0x0D and 0x0a	0x0D0A

5.7. General packet for information transmission

It is used to transmit all kinds of non positioning data at the terminal

Sample data: 79 79 00 20 94 0a 08 66 54 50 52 50 16 96 46 00 45 94 13 02 69 00 89 86 04 39
10 18 90 67 26 99 00 06 1b 25 0d 0a

		length	Explain in detail
Start bit		2	0x79 0x79
Package length		2	
Agreement number		1	0x94
information content	Information type (sub agreement No.)	1	00 external voltage 01 ~ 03 (customized) 04 terminal status synchronization 05 door status 08 self test parameters 09 positioning satellite information 0A iccid and other information To be added
	Data content	N	According to the different content of information, the content of simultaneous interpreting is different.
Information serial number		2	After starting up, the serial number of data sent each time will be automatically increased by 1
Error check		2	Crc-itu value from "package length" to "information serial number". If the receiving party has CRC error in the calculation of the received information, it will ignore it (discard this packet)
Stop bit		2	Fixed value, unified as 0x0D and 0x0a

When the type is 0A, this bit transmits the following information, which is a hexadecimal number 08 69 81 41 01 11 51 23 04 60 04 24 29 80 22 56 89 86 04 04 10 18 C0 00 22 55

IMEI	8	For example, if IMEI number is 0869 814101115123, the terminal ID is 0x08 0x69 0x81 0x41 0x01 0x11 0x51 0x23
IMSI	8	For example, if the IMSI number is 0460042429802256, the terminal ID is 0x04 0x60 0x04 0x24 0x29 0x80 0x22 0x56
ICCID	10	For example: if the ICCID number is 898604041018 c0002255, the terminal ID is 0x89 0x86 0x04 0x04 0x10 0x18 0xc0 0x00 0x22 0x55

5.6.1. Server response packet

format		length	describe	Examples
General package reply	Start bit	2	0x79 0x79	0x7878
	Package length	2		0x12
	Agreement number	1	0x94	0x80
	Sub agreement No	1	Consistent with the sending packet	0x0A
	Information serial number	2		0x00000001
	Error check	2		0x696363696423
	Stop bit	2	0x0D 0x0A	0x0001

5.7. Large file transfer (0x8d)

It is used to transmit large volume files such as voice files. The terminal sends data to the server using 0x8d protocol

5.8.1 Terminal sends to server

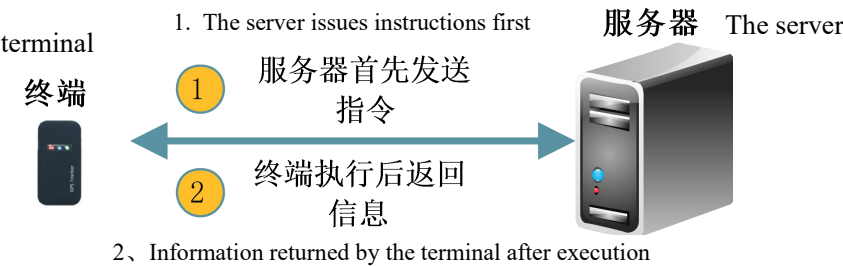
format		Length (Byte)	Explain in detail
informati on content	Start bit	2	0x79 0x79
	Package length	2	Length = protocol number + information content + information
	Agreement number	1	0x8D
	file type	1	0x00 recording file (listening) 0x01 recording file (voice control) 0x02 platform instruction request
	Total file length	4	Total length of transferred files
	File error check type	1	When the error check type is "00", CRC is used to check the transfer file When the error check type is "01", use MD5 to check the transfer file
	File error check code	N	When the error check type is "00", CRC is used to check the transmission file, and the length of the result is 2 bits When the error verification type is "01", use MD5 to verify the transmission file, and the length of the result is 16 bits
	Start position	4	Number of bytes at the start of transmission segmentation
	Current content length	2	The length of data after the start of transmission segmentation
	content	M	It' s a segmented packet
	Identification bit	N	The file type represents the type of the transfer file When the file type is 00 recording file (listening), this position is six bytes. The date and time when the transmission listening starts, and the encoding method is the same as the time format of the positioning package When the file type is 01 recording file (voice control), this position is six bytes. The date and time of transmission monitoring start, and the coding method is the same as the time format of positioning package When the file type is 02 platform instruction

			upload, the service ID is uploaded uniformly. The number of bytes continues to use 6 bytes. The first 4 bits are server flag bits and the last 2 bits are transmission serial numbers
	Information serial number	2	After starting up, the serial number of data sent each time will be automatically increased by 1
	Error check	2	Crc-itu value from "package length" to "information serial number". If the receiving party has CRC error in the calculation of the received information, it will ignore and discard the data packet (see Appendix A for the algorithm)
	Stop bit	2	Fixed value, unified as 0x0D and 0x0a

5.8.2 Server response packet

format	Length (Byte)	Explain in detail
信息内容	Start bit	2 0x79 0x79
	Package length	2 Length = protocol number + information content + information serial number + error verification
	Agreement number	1 0x8D
	exception status flag	1 Receive normal 0x00 receive error 0x01
	Information serial number	2 After starting up, the serial number of data sent each time will be automatically increased by 1
	Error check	2 Crc-itu value from "package length" to "information serial number". If the receiving party has CRC error in the calculation of the received information, it will ignore and discard the data packet (see Appendix A for the algorithm)
	Stop bit	2 Fixed value, unified as 0x0D and 0x0a

六. The server sends data packets to the terminal



6.1. Server send

format		length(Byte)
Start bit		2
Package length		1
Agreement number		1
information content	Instruction length	1
	Server flag bit	4
	Instruction content	M
	language	2
Information serial number		2
Error check		2
Stop bit		2

6.1.1. Start bit

See packet format 4.1 for details

6.1.2. Package length

See packet format 4.2 for details

6.1.3. Agreement number

Terminal sending protocol number: 0x80

6.1.4. Instruction length

Server flag bit + instruction content length

For example, in byte length, 0x0a means that the flag bit + instruction content occupies 10 bytes

6.1.5. Server flag bit

The terminal will return the received binary data in the return package

6.1.6. Instruction content

Represented by ASC II of string, the instruction content is compatible with SMS instruction(see 6.3-6.23 for details)

6.1.7. language

The current language bit of the terminal

Chinese: 0x00 0x01

English: 0x00 0x02

6.1.8. Information serial number

See packet format 4.5 for details

6.1.9. Error check

See packet format 4.6 for details

6.2. Terminal return (0x21)

format		Length (Byte)	Explain in detail
Start bit		2	0x79 0x79
Package length		2	0x00 0x09
Agreement number		1	0x21
informati on content	Instruction length	4	0x00 0x00 0x00 0x01
	Server flag bit	1	
	Instruction content	M	
	language	2	0x00 0x01
Information serial number		2	0xD9 0xDC
Error check		2	0x0D 0x0A

6.2.1 Start bit

Fixed value 0x79

6.2.2 Package length

Occupy 2 bytes

6.2.3 Agreement number

Use 0x21

6.2.4 server flag bit

The terminal will return the received binary data in the return package

6.2.5 content coding

0x01 ASC II code

0x02 utf16-be code

6.2.6 contents

Data to be sent.

6.2.7 information serial number

See packet format 4.5 for details

6.2.8 error verification

See packet format 4.6 for details

6.2.9 stop bit

See packet format 4.7 for details

6.3. Cut off oil and electricity

Function Description: cut off the vehicle oil and electricity control circuit

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

RELAY,1#

Terminal return

Successful return

Failure Return

6.4. Restore oil and electricity

Function Description: connect the vehicle oil and electricity control circuit

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

RELAY,0#

Terminal return

Successful return

Failure Return

Explain:

The following are various replies to the power cut-off. Success can be judged only by searching the success character.

RELAY, ERROR:103**OK,engine will be stopped after gps fixed or speed less than 20km/h****Already in the state of fuel supply cut off, the command is not running!****Already in the state of fuel supply to resume, the command is not running!****Cut off the fuel supply: Success!****Restore fuel supply:Success!**

6.5. Add SOS number

Function Description: add SOS number to receive alarm SMS and phone

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

```
SOS,A,NUM1,NUM2,NUM3#
```

Terminal return

Successful return

```
SOS_OK!SOS1:NUM1SOS2:NUM2SOS3NUM3
```

Failure Return

```
ERROR:XXX
```

6.6. Delete SOS number

Function Description: delete the SOS number receiving alarm SMS and phone

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

```
SOS,D,NUM1,NUM2,NUM3#
```

Terminal return

Successful return

```
SOS_OK!SOS1:NUM1SOS2:NUM2SOS3NUM3
```

Failure Return

```
ERROR:XXX
```

6.7. Set center number

Function Description: set the center number that can control the oil and electricity cut-off

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

```
CENTER,A,NUM#
```

Terminal return

Successful return

```
CENTER_OK
```

Failure Return

```
ERROR:XXX
```

6.8. Delete center number

Function Description: delete the center number that can control the oil and power cut-off

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

```
CENTER,D#
```

Terminal return

Successful return

```
CENTER_OK
```

Failure Return

```
ERROR:XXX
```

6.9. Turn on vibration alarm

Function Description: enable vibration alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

```
Server send
SENALM, ON,alarm mode#
```

```
Terminal return
Successful return
SENALM_OK
Failure Return
ERROR:XXX
```

Note: the alarm mode is 0: platform; 1: Platform + SMS; 2: Platform + SMS + phone call; 3 Platform + call;

6.10. Turn off vibration alarm

Function Description: turn off vibration alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

```
Server send
SENALM, OFF#
```

```
Terminal return
Successful return
SENALM_OK
Failure Return
ERROR:XXX
```

6.11. Open overspeed alarm

Function Description: enable overspeed alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

```
Server send
Speed, on, time, speed, alarm mode#
```

```
Terminal return
Successful return
SPEEDALM_OK
Failure Return
ERROR:XXX
```

Note: the alarm mode is 0: platform; 1: Platform + SMS;

6.12. Turn off overspeed alarm

Function Description: close overspeed alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

```
Server send
```

SPEED, OFF#

Terminal return

Successful return

SPEEDALM_OK

Failure Return

ERROR:XXX

6.13. Turn on the power-off alarm

Function Description: turn on the power-off alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

Powerarm, on, alarm mode#

Terminal return

Successful return

POWERALM_OK

Failure Return

ERROR:XXX

Note: the alarm mode is 0: platform; 1: Platform + SMS; 2: Platform + SMS + phone call; 3 Platform + call;

6.14. Turn off the power-off alarm

Function Description: turn off the power-off alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

POWERALM, OFF#

Terminal return

Successful return

POWERALM_OK

Failure Return

ERROR:XXX

6.15. Alarm on displacement

Function Description: enable displacement alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

Moving, no, displacement radius, alarm mode#
--

Terminal return

Successful return

MOVING_OK

Failure Return

ERROR:XXX

Note: the alarm mode is 0: platform; 1: Platform + SMS; 2: Platform + SMS + phone call; 3 Platform + call;

Displacement radius: 100 ~ 1000

6.16. Close displacement alarm

Function Description: close displacement alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

MOVING ,OFF#

Terminal return**Successful return**

MOVING_OK

Failure Return

ERROR:XXX

6.17. Turn on low power alarm

Function Description: turn on low power alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

Batalm, on, alarm mode#

Terminal return**Successful return**

BATALM_OK

Failure Return

ERROR:XXX

Note: the alarm mode is 0: platform; 1: Platform + SMS;

6.18. Turn off low power alarm

Function Description: turn off low power alarm

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

BATALM, OFF#

Terminal return**Successful return**

BATALM_OK

Failure Return

ERROR:XXX

6.19. Open fence alarm

Function Description: open the electronic fence

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

```
Server send
Fence, on, 0, center latitude, center longitude, fence radius, x, alarm mode#
For example: fence, on, 0, N1 2971,E103. 822349,61,IN,0#
```

```
Terminal return
Successful return
FENCE_OK
Failure Return
ERROR:XXX
```

Note: the alarm mode is 0: platform; 1: Platform + SMS;

X=IN/OUT; In: alarm when entering the fence, out: alarm when leaving the fence. If it is blank, the alarm will be given when entering / leaving the fence; The default is that the alarm will be given when entering / leaving the fence;

6.20. Close fence alarm

Function Description: open the electronic fence

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

```
Server send
FENCE, OFF#
```

```
Terminal return
Successful return
FENCE_OK
Failure Return
ERROR:XXX
```

6.21. Control equipment restart

Function Description: restart the device after 1 minute

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

```
Server send
RESET#
```

```
Terminal return
Successful return
The terminal will restart after 1 minute!
Failure Return
ERROR:XXX
```

6.22. Recording instruction

Function Description: after receiving the instruction, the terminal immediately starts recording

and uploads it through 0x8d instruction

The sending and returning strings will be converted into ASCII to generate instruction contents in the instance

Server send

LY,XX#

XX represents the recording time, 10-300 (in seconds), 999 means to turn on continuous recording, and 0 means to turn off continuous recording

Terminal return

Successful return

When XX is 0 or 999: CLY_OK

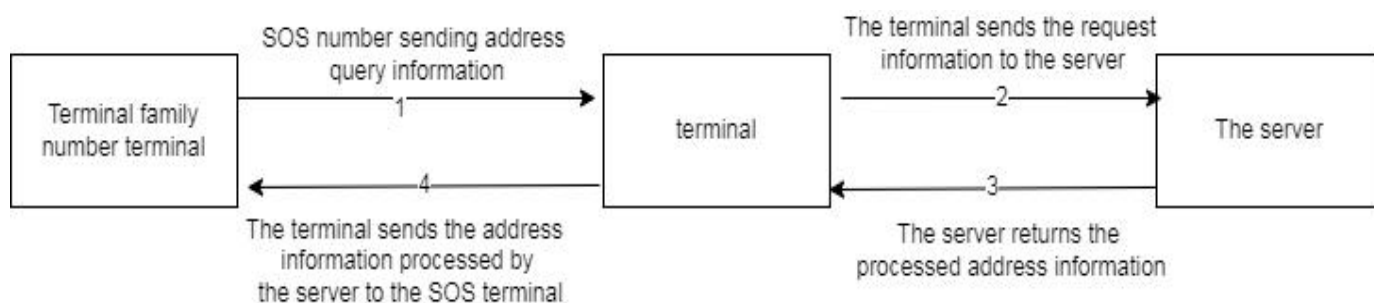
When XX is 10-300: ly_OK:Start recording.

Failure Return

ERROR:XXX

6.23. GPS, telephone number inquiry address information package (0x2a)

After receiving the address query instruction sent from the SOS number, the terminal starts positioning and reports the GPS information to the server through the 0x2a instruction. The server returns the address information corresponding to longitude and latitude, and the terminal sends the received address information to the SOS number.



6.23.1. The terminal sends the query address information packet to the server

The information content is basically the same as the format mentioned in the previous GPS information content, and a telephone number for querying the address is added.

format			Length (Byte)
Start bit			2
Package length			1
Agreement number			1
informat ion content	Date time		6
	GPS informat ion	Length of GPS information and number of satellites involved in positioning	1
		latitude	4
		longitude	4
		speed	1
		Heading and status	2

	Telephone number	21
	language	2
Information serial number		2
Error check		2
Stop bit		2

6.7.1.1. Start bit

See packet format 4.1 for details

6.7.1.2. Package length

See packet format 4.2 for details

For example, in byte length, 0x2e means that the instruction content occupies 46 bytes

6.7.1.3. Agreement number

Use **0x2A** .

6.7.1.4. Date time

See packet format 5.2.1.4 for details

6.7.1.5. Length of GPS information and number of satellites involved in positioning

See packet format 5.2.1.5 for details

6.7.1.6. latitude

See packet format 5.2.1.6 for details

6.7.1.7. longitude

See packet format 5.2.1.7 for details

6.7.1.8. speed

See packet format 5.2.1.8 for details

6.7.1.9. course

See packet format 5.2.1.9 for details

6.7.1.10. Telephone number

The SOS telephone number requesting address query is converted through ASCII, and 0 is added to the right of less than 21 digits

6.7.1.11. language

The current language bit of the terminal

Chinese: 0x00 0x01

English: 0x00 0x02

6.7.1.12. Information serial number

See packet format 4.5 for details

6.7.1.13. Error check

See packet format 4.6 for details

6.7.1.14. Stop bit

See packet format 4.7 for details

6.23.2. Server return address information

The server combines the telephone numbers in the 0x2a information content into the following contents for transmission

Address & & address content & & phone number##

Note: the Chinese address content is distributed in Unicode code.

Reply to the Chinese address or English address according to the extension instruction request,

and the reply packet is inconsistent.

6.23.2.1. Chinese reply (0x17)

The Chinese reply packet is as follows:

format				Length (Byte)	示例	
Instruction package sent by the server to the terminal (15 + m + n byte)	Start bit			2	0x7878	
	Data bit length			2	44+M	
	Agreement number			1	0x17	
	informat ion content	Instruction length		2	38+M	
		Server flag bit		4	0x00000001	
		Instruct ion content	ADDRESS		7	0x41444452455353
			&&		2	0x2626
			Address content		M	0x624059044F4D7F.. .
			&&		2	0x2626
			Telephone number		21	0x00000000...
			##		2	0x2323
		Information serial number			2	0x0010
	Check bit			2	0x6AB1	
Stop bit			2	0x0D0A		

Request Chinese address reply agreement No.: 0x17

Instruction content: Address & & address content & & phone number ## (address, & &, ## are fixed strings)

The Chinese address content is distributed in Unicode.

Example of reply to Chinese address information:

```

seven thousand eight hundred and seventy-eight // Start bit
eighty-four // Data length
seventeen // Reply agreement No
7E // The instruction length is the length of the transmitted content
information
00000001 // Server flag bit
forty-one trillion and four hundred and forty-four billion four hundred and fifty-two million
four hundred and fifty-five thousand three hundred and fifty-three // ADDRESS
two thousand six hundred and twenty-six //&& Separator
624059044f4d7f6e0028 / / send the Chinese position in Unicode
004C004200530029003A
5E7F4E1C77015E7F5DDE
5E0282B190FD533AFF17
FF15FF144E6190530028
004E00320033002E0033
00390035002C00450031
00310032002E00390038

```

```

0038002996448FD1
two thousand six hundred and twenty-six //&& Separator
31333731303831393133350000000000000000 / / phone number
two thousand three hundred and twenty-three //## Content information Terminator
0106 // serial number
three thousand eight hundred and twenty-five // Check bit
0D0A // Stop bit

```

6.23.2.2. English reply

Considering the long English or other foreign addresses, one data bit is not enough, and it is increased to 2 bytes. be careful:

Among them, the length of data bits corresponding to the protocol number of back address information is changed to 2.

format				Length (Byte)	示例
Instruction package sent by the server to the terminal (15 + m + n byte)	Start bit			2	0x7979
	Data bit length			2	45+M
	Agreement number			1	0x97
	informat ion content	Instruction length		2	38+M
		Server flag bit		4	0x00000001
		Instruct ion content	ADDRESS	7	0x4144445245535 3
			&&	2	0x2626
			Address content	M	0x534F53284C29. ..
			&&	2	0x2626
			Telephone number	21	0x00000000...
			##	2	0x2323
		Information serial number		2	0x0011
		Check bit		2	0x1A32
		Stop bit		2	0x0D0A

Request English address reply agreement No.: 0x97

Instruction content: Address && && address content && && phone number ## (address, && &&, ## are fixed strings)

Example reply to English address information example:

```

7878 / / start bit
00d1 / / data length
97 / / reply to agreement No
00CA / / the instruction length is the length of the sent content information
00000001 / / server flag bit
41444452455353 //ADDRESS
2626 / / && && separator

```

0053004f00530028004c // the English position is sent in Unicode
0029003A005300680069
006D0069006E00200046
0061006900720079006C
0061006E006400200057
00650073007400200052
0064002C004800750069
006300680065006E0067
002C004800750069007A
0068006F0075002C0047
00750061006E00670064
006F006E00670028004E
00320033002E00310031
0031002C004500310031
0034002E003400310031
0029004E006500610072
00620079
2626 // & separator
31323532303133373930373430353100000000 // phone number
2323 // ## content information Terminator
0007 // serial number
72b5 // check bit
0d0a // stop bit

七. Attachment a crc-itu table look up algorithm C language code fragment

Crc-itu table lookup algorithm C language code fragment

```
static const U16 crctab16[] =
{
    0X0000, 0X1189, 0X2312, 0X329B, 0X4624, 0X57AD, 0X6536, 0X74BF,
    0X8C48, 0X9DC1, 0XAF5A, 0XBED3, 0XCA6C, 0XDBE5, 0XE97E, 0XF8F7,
    0X1081, 0X0108, 0X3393, 0X221A, 0X56A5, 0X472C, 0X75B7, 0X643E,
    0X9CC9, 0X8D40, 0XBFDB, 0XAE52, 0XDAED, 0XCB64, 0XF9FF, 0XE876,
    0X2102, 0X308B, 0X0210, 0X1399, 0X6726, 0X76AF, 0X4434, 0X55BD,
    0XAD4A, 0XBCC3, 0X8E58, 0X9FD1, 0XEB6E, 0XFAE7, 0XC87C, 0XD9F5,
    0X3183, 0X200A, 0X1291, 0X0318, 0X77A7, 0X662E, 0X54B5, 0X453C,
    0XBDCB, 0XAC42, 0X9ED9, 0X8F50, 0XFBF7, 0XEA66, 0XD8FD, 0XC974,
    0X4204, 0X538D, 0X6116, 0X709F, 0X0420, 0X15A9, 0X2732, 0X36BB,
    0XCE4C, 0XDFC5, 0XED5E, 0XFCDD, 0X8868, 0X99E1, 0XAB7A, 0XBAF3,
    0X5285, 0X430C, 0X7197, 0X601E, 0X14A1, 0X0528, 0X37B3, 0X263A,
    0XDECD, 0XCF44, 0XFDFF, 0XEC56, 0X98E9, 0X8960, 0XBBFB, 0XAA72,
    0X6306, 0X728F, 0X4014, 0X519D, 0X2522, 0X34AB, 0X0630, 0X17B9,
    0XEF4E, 0XFEC7, 0XCC5C, 0XDDD5, 0XA96A, 0XB8E3, 0X8A78, 0X9BF1,
    0X7387, 0X620E, 0X5095, 0X411C, 0X35A3, 0X242A, 0X16B1, 0X0738,
    0XFFCF, 0XEE46, 0XDCDD, 0XCDD5, 0XB9EB, 0XA862, 0X9AF9, 0X8B70,
    0X8408, 0X9581, 0XA71A, 0XB693, 0XC22C, 0XD3A5, 0XE13E, 0XF0B7,
    0X0840, 0X19C9, 0X2B52, 0X3ADB, 0X4E64, 0X5FED, 0X6D76, 0X7CFF,
    0X9489, 0X8500, 0XB79B, 0XA612, 0XD2AD, 0XC324, 0XF1BF, 0XE036,
    0X18C1, 0X0948, 0X3BD3, 0X2A5A, 0X5EE5, 0X4F6C, 0X7DF7, 0X6C7E,
    0XA50A, 0XB483, 0X8618, 0X9791, 0XE32E, 0XF2A7, 0XC03C, 0XD1B5,
    0X2942, 0X38CB, 0X0A50, 0X1BD9, 0X6F66, 0X7EEF, 0X4C74, 0X5DFD,
    0XB58B, 0XA402, 0X9699, 0X8710, 0XF3AF, 0XE226, 0XD0BD, 0XC134,
    0X39C3, 0X284A, 0X1AD1, 0X0B58, 0X7FE7, 0X6E6E, 0X5CF5, 0X4D7C,
    0XC60C, 0XD785, 0XE51E, 0XF497, 0X8028, 0X91A1, 0XA33A, 0XB2B3,
    0X4A44, 0X5BCD, 0X6956, 0X78DF, 0X0C60, 0X1DE9, 0X2F72, 0X3EFB,
    0XD68D, 0XC704, 0XF59F, 0XE416, 0X90A9, 0X8120, 0XB3BB, 0XA232,
    0X5AC5, 0X4B4C, 0X79D7, 0X685E, 0X1CE1, 0X0D68, 0X3FF3, 0X2E7A,
    0XE70E, 0XF687, 0XC41C, 0XD595, 0XA12A, 0XB0A3, 0X8238, 0X93B1,
    0X6B46, 0X7ACF, 0X4854, 0X59DD, 0X2D62, 0X3CEB, 0X0E70, 0X1FF9,
    0XF78F, 0XE606, 0XD49D, 0XC514, 0XB1AB, 0XA022, 0X92B9, 0X8330,
    0X7BC7, 0X6A4E, 0X58D5, 0X495C, 0X3DE3, 0X2C6A, 0X1EF1, 0X0F78,
};
```

/Calculates a 16 bit CRC for a given length of data.

U16 GetCrc16(const U8* pData, int nLength)

```
{
    U16 fcs = 0xffff;           // initialization
    while(nLength>0){
        fcs = (fcs >> 8) ^ crctab16[(fcs ^ *pData) & 0xff];
        nLength--;
        pData++;
    }
    return ~fcs;               // Reverse}
}
```


			005,0#
10	Turn on power-off alarm setting	POWERALM, A, M, T1, T2, # A=ON M=0~2; 0 GPRS only, 1 SMS + GPRS, 2 GPRS + SMS + call T1 = 2 ~ 60s; Power off detection time T2 = 1-3600 seconds; Minimum charging time	
11	Power off detection time	POWERALM,OFF#	
12	Query power failure alarm status	POWERALM#	
13	Turn on low power alarm setting	BATALM, A, M# A=ON M=0~1 0 stands for GPRS only, 1 stands for SMS + GPRS,	
15	Turn off low power alarm setting	BATALM,OFF#	
16	Query low power alarm status	BATALM#	
17	Enable displacement alarm setting	MOVING, A, R, M# A=ON R=100~1000; Displacement radius M=0~2; 0 仅 GPRS, 1 SMS+GPRS; 2 GPRS+SMS+CALL	
18	Close displacement alarm	MOVING,OFF#	
19	Query displacement setting status	MOVING#	
20	sound recording	LY,60#	